Is In-Context Learning in Large Language Models Bayesian? A Martingale Perspective

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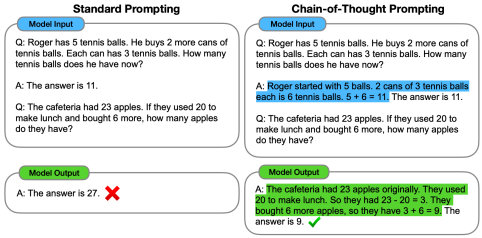
Presented by Qin Weng

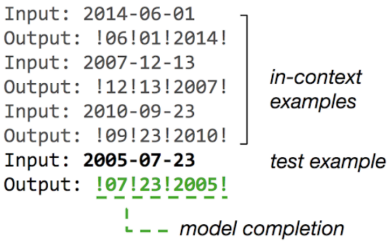
1. Background: In-Context Learning (ICL)

Definition. Given a pretrained LLM p_M and an in-context dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\} = z_{1:n}$ written into the prompt, the model predicts *without updating its weights*, either a label or a new sample:

$$p_M(Y_{n+1} \mid X_{n+1} = x_{n+1},\ Z_{1:n} = z_{1:n}) \qquad \text{or} \qquad p_M(X_{n+1}, Y_{n+1} \mid Z_{1:n} = z_{1:n}).$$

Common uses: Few-shot classification, chain-of-thought prompting, and synthetic data generation.





An ICL Example: Imputing the Rest of a Trial

Setting. A two-arm randomized controlled trial (RCT) with 100 patients. At the interim analysis, 50 patients have been observed.

Procedure. Write the 50 patients into the prompt (context) and let the LLM generate the remaining patients one by one:

$$\hat{z}_{n+k} \sim p_M(\cdot \mid z_{1:n}, \hat{z}_{n+1:n+k-1}), \qquad k = 1, \ldots, 50.$$

Repeat J times, and decide to continue the trial if the final trial outcome (100 patients) shows success.
$\rightarrow$ Mirrors Bayesian adaptive trial designs.

Can we trust a conclusion reached by LLM's prediction using ICL under this procedure?

What a Trustworthy Imputation Requires

1. **Requirement 1: Unambiguous predictions** *(data perspective)*
   Randomization makes the patients **exchangeable**. The model should treat all orders alike, so imputing patient #51 or #100 first must not change the conclusion.

2. **Requirement 2: Uncertainty decomposition** *(decision perspective)*
   We need to know how much uncertainty 50 more patients would resolve to decide whether to continue. The decomposition of epistemic vs. aleatoric uncertainty is meaningful only if average belief does not drift: the **martingale property**.

Does ICL in LLMs meet these requirements?

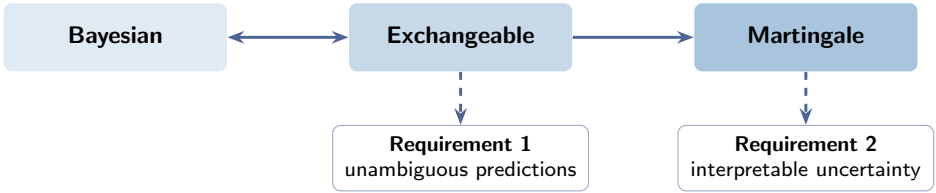
- A popular claim: ICL in LLM as approximately **Bayesian inference**.
- If true, both requirements holds.

Hypothesis and Roadmap

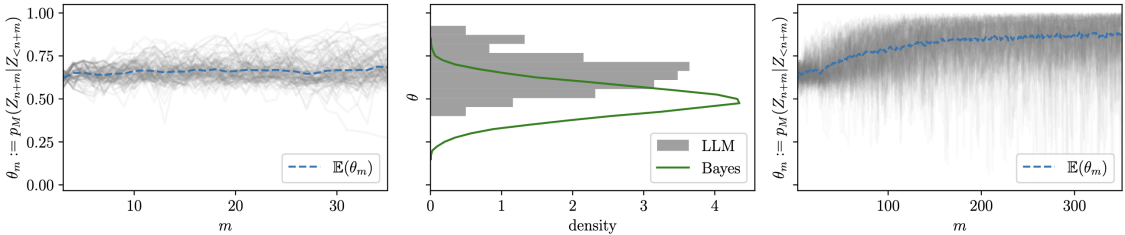
Hypothesis

For exchangeable (e.g. i.i.d.) in-context data, ICL in LLMs is (approximately) Bayesian.

- Neither the Bayesian claim (the prior is unobservable) nor exchangeability (all permutations) can be tested directly.
- The **martingale property** is necessary for both, and it can be checked from the model's own imputed paths.



Results Preview: ICL in LLMs is not Bayesian



- **Left:** on short paths, the mean forecast stays flat.
- **Centre:** the posterior implied by these paths differs from the reference Bayesian posterior.
- **Right:** on long paths, the mean drifts upward, the model is persuaded by its own guesses.

The LLM does not behave like a Bayesian learner.

2. Theoretical Analysis

We build the chain **Bayesian** $\iff$ **exchangeable** $\Rightarrow$ **martingale** in three steps.

1. **Martingale & Exchangeability**
   Show exchangeable $\Rightarrow$ martingale.
   $\rightarrow$ Requirement 1: unambiguous predictions

2. **Martingale & Uncertainty**
   Show martingale posterior $\Rightarrow$ uncertainty decomposition.
   $\rightarrow$ Requirement 2: uncertainty decomposition

3. **Link Martingale to Bayes**
   Show Bayesian $\iff$ exchangeable.
   $\rightarrow$ Why a violation of the martingale rules out Bayes.

The Martingale Property

Definition 1 (Martingale property)

The predictive distributions for $\{Z_i\}$ satisfy the martingale property if, for all integers $n, k > 0$ and realizations $\{z, z_{1:n}\}$,

$$p_M(Z_{n+1} = z \mid Z_{1:n} = z_{1:n}) \;=\; p_M(Z_{n+k} = z \mid Z_{1:n} = z_{1:n}). \tag{1}$$

The unobserved samples in between are averaged out using the model's own forecasts:

$$p_M(Z_{n+k} = z \mid Z_{1:n}) = \mathbb{E}_{Z_{n+1:n+k-1} \sim p_M(\cdot \mid Z_{1:n})}\big[\, p_M(Z_{n+k} = z \mid Z_{1:n+k-1}) \,\big].$$

- The Z_i are conditionally identically distributed.
- Imputing samples must not change the forecast on average.

Exchangeability

Exchangeability

$\{Z_i\} \sim p_M$ is exchangeable if for all $\ell \in \mathbb{N}$ and permutations σ of $\{1, \ldots, \ell\}$,

$$p_M(Z_1, \ldots, Z_\ell) = p_M(Z_{\sigma(1)}, \ldots, Z_{\sigma(\ell)}). \tag{3}$$

- Martingale property constrains the **marginal** of each future position given the past.
- Exchangeability constrains the **joint** distribution of the whold sequence.

Exchangeable $\Rightarrow$ Martingale

How Strong Is Each Constraint?

Example: Draw from balls from a box. After each draw:

Rule	i.i.d.	Exchangeable	Martingale
Put the ball back, add nothing	✓	✓	✓
Add one same-color ball (a Bayesian learner)	×	✓	✓
Add 1 same-color ball, then 10, then ... (later draws weigh more: order matters)	×	×	✓
Always add an extra red ball (belief drifts toward red)	×	×	×

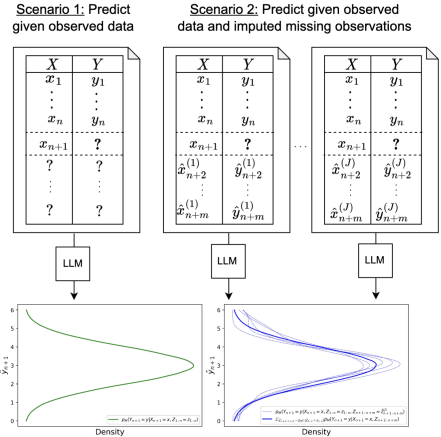
i.i.d. $\subset$ exchangeable $\subset$ martingale

Martingale Property Is Necessary for Unambiguous Predictions

With $Z = (X, Y)$, if martingale holds, then:

$$\underbrace{p_M(Y_{n+1} \mid x,\, Z_{1:n})}_{\text{predict directly}} = \underbrace{\mathbb{E}_{Z_{n+1:n+k-1}}\, p_M(Y_{n+k} \mid x,\, Z_{1:n+k-1})}_{\text{impute, predict, average}}.$$

- Imputed samples carry no new information.
- A model that changes its forecast by querying itself shows an **introspective hallucination**.



Violating martingale ⇒ not exchangeable ⇒ predictions depend on the imputation order

From Forecasts to a Parameter

A black-box LLM has no explicit parameter. Idea: build one from the model's own **forecasts**.

1. **Forecasts form a martingale.** Let $\theta_n := p_M(Z_{n+1} = \cdot \mid Z_{1:n}) \in \mathbb{R}^A$, the forecast after n samples. By the tower property, $\{\theta_n\}$ is a **martingale**:

$$\underbrace{p_M(Z_{n+1} = z \mid Z_{1:n}) = p_M(Z_{n+k} = z \mid Z_{1:n}) \;\; \forall k}_{\text{Eq. (1)}} \iff \mathbb{E}[\theta_{n+1} \mid Z_{1:n}] = \theta_n.$$

2. **Forecasts converge.** A bounded martingale converges (Doob's theorem):

$$\theta_n \to \theta_\infty \quad \text{a.s.}, \qquad \theta_n = \mathbb{E}[\theta_\infty \mid Z_{1:n}].$$

- θ_∞ is the forecast the model would make after infinitely many samples, as a "parameter".
- θ_n is the average of all the forecasts the model could end up with.

The Martingale Posterior

Writing $\theta_n = \mathbb{E}[\theta_\infty \mid Z_{1:n}]$ as an integral over θ_∞ gives the same form as Bayesian prediction:

Martingale representation vs. Bayesian posterior predictive

$$p_M(Z_{n+1} = \cdot \mid Z_{1:n}) = \int \underbrace{p(\theta_\infty \mid Z_{1:n})}_{\text{martingale posterior}} \underbrace{p_{\theta_\infty}(Z = \cdot)}_{\text{acts as likelihood}} \, d\theta_\infty \tag{4}$$

$$p(Z_{n+1} = \cdot \mid Z_{1:n}) = \int \underbrace{p(\theta \mid Z_{1:n})}_{\text{posterior}} \underbrace{p(Z = \cdot \mid \theta)}_{\text{likelihood}} \, d\theta \tag{5}$$

- θ_∞ plays the role of the parameter θ.
- If p_M is Bayesian, $p(\theta_\infty \mid Z_{1:n})$ equals the Bayesian posterior.
- $p(\theta_\infty \mid Z_{1:n})$ can be sampled by imputing a long path and read off the final forecast.

Martingale Property Enables Uncertainty Decomposition

Eq. (4) shows that the predictive uncertainty has two sources:

$$p_M(Z_{n+1} = \cdot \mid Z_{1:n}) = \int \underbrace{p(\theta_\infty \mid Z_{1:n})}_{\text{epistemic}} \underbrace{p_{\theta_\infty}(Z = \cdot)}_{\text{aleatoric}} \, d\theta_\infty \tag{4}$$

- **Epistemic:** reduced by more real data (measured by imputed paths).
- **Aleatoric:** remains even if θ_∞ were known.

Why the martingale property is needed:

The spread of future beliefs measures learning only if beliefs move *because of* information, otherwise the spread mixes learning with the model's own drift.

Bayesian $\iff$ Exchangeable $\Rightarrow$ Martingale

For i.i.d. data, a correctly specified Bayesian model treats observations as **conditionally i.i.d.** given a parameter:

$$p_M(Z_{1:n} = z_{1:n}) = \int \Big(\prod_{i=1}^{n} p_M(Z = z_i \mid \theta) \Big) \pi(\theta)\, d\theta, \qquad \forall n. \tag{7}$$

- **Bayesian $\Rightarrow$ exchangeable:** the product in Eq. (7) is invariant to permuting $z_{1:n}$.
- **Exchangeable $\Rightarrow$ Bayesian:** by de Finetti's theorem, any exchangeable p_M admits Eq. (7).

Conclusion

- For ICL on i.i.d. data: Bayesian $\iff$ exchangeable $\Rightarrow$ martingale property.
- Any ICL system that violates the martingale property **cannot be Bayesian**.

3. Diagnostic: What Should Be Tested?

Testing the martingale property on an LLM raises three problems.

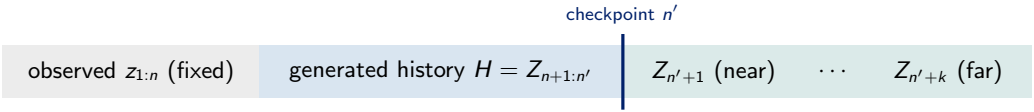
Problem		Design choice
Too many implications: the property fixes every conditional moment.	→	Test the two that matter: the **mean** and the **covariance**.
LLMs only approximate: error accumulates along long imputed paths.	→	Use **short paths** ($m \in \{n/2, 2n\}$) and remove outlier paths.
Small deviations are harmless: a perfect match is neither expected nor needed.	→	Look for **clear** violations: tolerate $o(1)$ error in the mean, $o(1/n)$ in the covariance.

Diagnostic tests:

- **Martingale property** (Requirement 1): mean drift $T_{1,g}$ and covariance $T_{2,k}$.
- **Uncertainty scaling** (Requirement 2): spread of the martingale posterior T_3.

Proposition 1: From Conditional to Weighted Averages

At every checkpoint n', Definition 1 requires $\mathbb{E}\left[g(Z_{n'+k}) - g(Z_{n'+1}) \,\middle|\, Z_{1:n},\, Z_{n+1:n'}\right] = 0$ for every generated history $H := Z_{n+1:n'}$.



- Each path has its own H, so the conditional means cannot be estimated.
- Proposition 1 averages over all paths, with a weight h that groups paths by their H.

Proposition 1

A sequence $\{Z_{n+1:n+m}\} \sim p_M(\cdot \mid Z_{1:n})$ satisfies the martingale property iff: for all $n', k \in \mathbb{N}$ and integrable functions g, h:

$$\mathbb{E}\Big(\big(g(Z_{n'+k}) - g(Z_{n'+1})\big)\, h(Z_{n+1:n'}) \,\Big|\, Z_{1:n}\Big) = 0. \tag{8}$$

Checking Martingale Property

If $\{Z_i\}$ satisfies the martingale property, then for all integers $n' > n$, $k > 0$:

- **(i) Mean** $(h \equiv 1,\ g(z) = z)$:

$$\mathbb{E}\big(Z_{n+1} \mid Z_{1:n}\big) = \mathbb{E}\big(Z_{n+k} \mid Z_{1:n}\big).$$

  For a Bayesian model with $\mathbb{E}[Z \mid \theta] = \theta$, both sides equal the **posterior mean** $\mathbb{E}(\theta \mid Z_{1:n})$.

- **(ii) Covariance** $(h = Z_{n'},\ g(z) = z)$:

$$\mathbb{E}\big((Z_{n'+k+1} - Z_{n'+1})\, Z_{n'}^{\top} \mid Z_{1:n}\big) = 0.$$

  For a Bayesian model it equals $\mathbb{E}(\theta\theta^{\top} \mid Z_{1:n})$, which with (i) fixes the **posterior covariance**.

Both are estimated by averages over J imputed paths $(T_{1,g},\ T_{2,k})$ and compared with a bootstrap band from a reference Bayesian model.

Does Epistemic Uncertainty Shrink at the Bayesian Rate?

The martingale property allows to measure epistemic uncertainty, which should decrease with more observed samples.

Fact 1: epistemic uncertainty = estimation error

For a Bayesian model with posterior mean $\bar{\theta}_n$,

$$\mathbb{E}_{\theta_0 \sim \pi,\, z_{1:n} \sim p(\cdot|\theta_0)}\, \mathbb{E}_{\theta \sim \pi(\theta|z_{1:n})} \|\theta - \bar{\theta}_n\|^2 = \mathbb{E}_{\theta_0 \sim \pi,\, z_{1:n} \sim p(\cdot|\theta_0)} \|\theta_0 - \bar{\theta}_n\|^2. \tag{9}$$

- The estimation error is $O(d/n)$, where n is data size and d is dimension of the parameter.
- **Statistic:** on each imputed path $\{z_{n+i}^{(j)}\}_{i=1}^m$, fit the MLE under the known likelihood form p_θ,

$$\hat{\theta}_m^{(j)} := \arg\max_{\theta \in \Theta} \sum_{i=1}^{m} \log p_\theta\big(z_{n+i}^{(j)}\big), \qquad T_3 = Q_{0.75}\big(\{\hat{\theta}_m^{(j)}\}_{j=1}^J\big) - Q_{0.25}\big(\{\hat{\theta}_m^{(j)}\}_{j=1}^J\big),$$

the spread of the MLEs across path approximates the martingale posterior.

Experimental Setup

Three synthetic datasets:

Task	Data-generating process
Bernoulli	$Z_i \sim \mathrm{Bern}(\theta)$, $\theta \in \{0.3, 0.5, 0.7\}$
Gaussian	$Z_i \sim \mathcal{N}(\theta, 1)$, $\theta \in \{-1, 0, 1\}$
Clinical text	$X_i \sim \mathrm{Bern}(0.5)$ (symptom), $Y_i \mid X_i \sim \mathrm{Bern}(0.3 + 0.4X_i)$ (disease)

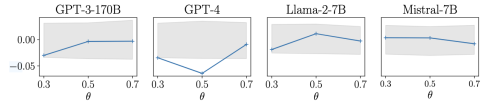
- **Models:** Llama-2-7B, Mistral-7B, GPT-3 (2.7B, 170B), GPT-3.5, GPT-4.
- **Prompt:** states the likelihood family and that observations are i.i.d.:

  ```
  "You will make predictions for a novel disease. The observed dataset contains records for multiple subjects which are assumed to be independent and identically distributed. For each subject there are two binary variables, indicating fever and disease diagnosis, respectively. Output your prediction for the disease diagnosis of the next subject.Id: 0Fever: YDiagnosis: N..."
  ```

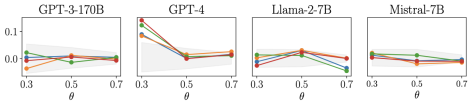
- Each LLM and the reference Bayesian model go through the same sampling procedure.

Bernoulli Experiments

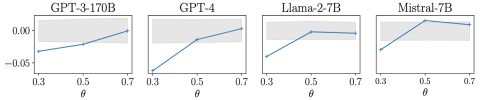
(a) $T_{1,g}$ for $g(z) = z, n = 50, m = 25$



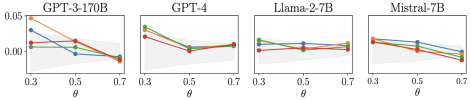
(b) $T_{2,k}$ for $k \in \{2, 3, 4, 5\}, n = 50, m = 25$



(c) $T_{1,g}$ for $g(z) = z, n = 50, m = 100$

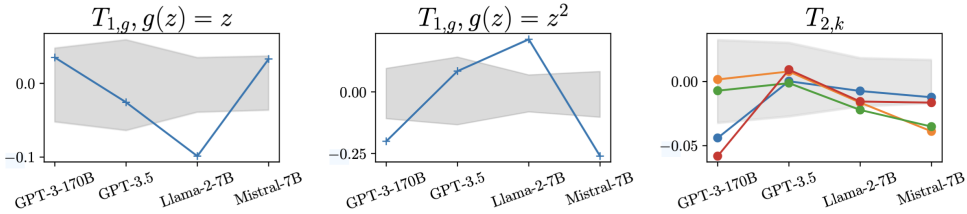


(d) $T_{2,k}$ for $k \in \{2, 3, 4, 5\}, n = 50, m = 100$



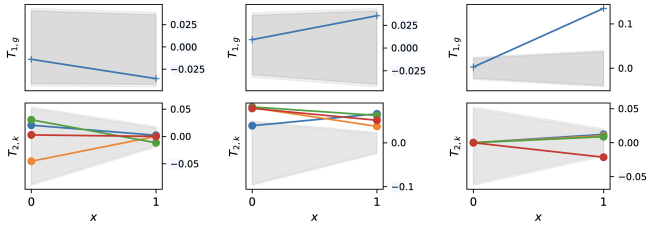
- Short vs. long paths: most models pass short paths but deviate increasingly at longer paths as per-step errors accumulate.
- GPT-3-170B $>$ GPT-3.5 $>$ GPT-4; the two weakest are instruction-tuned.

Gaussian Experiments



- $Z_i \sim \mathcal{N}(-1, 1)$, $n = 100$, short paths $m = n/2$.
- Continuous values (spans several tokens) break the martingale property already on short paths.

Clinical Text Experiments

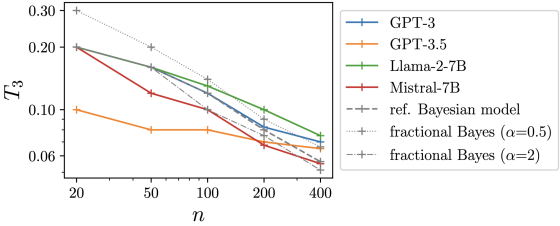


(a) GPT-3-170B (b) GPT-3.5 (c) GPT-4

- $n = 80$, $m = 40$. The checks test the conditional predictive $p_M(Y \mid X, Z_{1:i})$.
- Violations persist in conditional prediction from natural-language records.

Scaling of Epistemic Uncertainty

On Bernoulli data ($\theta = 0.5$, $m = n/2$), where all models roughly satisfy the martingale property, track how T_3 shrinks with n, against the reference Bayesian model and fractional posteriors.



- **Consistent with the martingale checks:** GPT-3-170B and Mistral-7B, the best on the martingale checks, are also the closest to the Bayesian rate.
- **Martingale holds, uncertainty still fails:** in this setting all models roughly satisfy the martingale property, yet Llama-2-7B and GPT-3.5 shrink uncertainty more slowly than Bayes, and GPT-3.5 is overconfident at small n.

Summary and Limitations

Takeaways

- For exchangeable data, Bayesian $\iff$ exchangeable $\Rightarrow$ martingale property.
- The martingale property makes predictions invariant to the imputation order, and defines a martingale posterior that separates epistemic from aleatoric uncertainty.
- LLMs violate of the martingale property, and deviate from a Bayesian scaling behaviour of uncertainty, falsifying the hypothesis that ICL is Bayesian.

Limitations. Deliberately small, synthetic datasets.

Discussion

Worth reading? Yes.

- **Solid theory**: links Bayesian, exchangeability and martingale property into one testable chain, and later work builds on it.
- **Practical value**: Checks Bayesian ICL through self-consistency of imputations, which matters wherever uncertainty drives decisions.

Worth implementing? Maybe.

- **Hard to scale up:** each setting needs many long imputed paths and massive sampling.
- **Still useful** as a black-box sanity check before trusting LLM imputation or uncertainty in small, well-specified tasks.